

Sachin Konan

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EDUCATION

Doctor of Philosophy, Princeton University

Fall 2025 -

Computer Science; Advisor: [Zhuang Liu](#)

Current

Topic: *Building and Evaluating Automated Scientific Research Agents*

- RL: Efficient post-training for long-context, tool-heavy environments
- Reward Models: Designing verifiers that correctly attribute scientific impact

Bachelor of Science, Georgia Institute of Technology

2018 - 2022

Computer Science; Threads in Theory and Artificial Intelligence

Research Advisor: [Matthew Gombolay](#)

RESEARCH EXPERIENCE

Two Sigma - Metadata

Fall 2022 -

Current

- Auto-Featurization: Building LLMs that can generate and “plan” data pipelines
- Semantic Types: Developed a framework to dedupe pipelines with entity detection ([In NAACL 2024](#))
- Finetuned open-source LLMs for finance-specific detection and code generation
- Synthetic Directed Graphs: Theorized “[Merge-Split](#)”, transformations that retain spectral characteristics of directed graphs while ensuring subgraph privacy
- Compression: Engineered 10x reduction in Two Sigma’s Data Lineage Graph

Georgia Institute of Technology - Research Assistant (CORE Robotics Laboratory)

Fall 2020 -

Spring 2022

- *InfoPG*: Developed an information-theoretic objective encouraging collaboration in multi-agent, collaborative RL games ([In ICLR 2022](#))
- *ConDT*: Improved the architecture of Decision Transformer to discriminate between state-action embeddings based on return ([In CoRL 2023](#))
- For *InfoPG/ConDT*, implemented multi-GPU training on the lab’s cluster using Ray

Meta - Fundamental AI Research (FAIR)

Fall 2021

- Open World Detection for One-Stage Detection ([On arXiv](#))

Intel - Deep Learning Group

Summer -

- Improved runtime of ArchBench (DNN hardware simulator) from 2 hrs to 4 mins with tensor caching and parallelization

Fall 2019

PUBLICATIONS

4. Guohui Guan, **Sachin Konan**, Larry Rudolph, Chang Ge. “[SemForest: Semantic-Aware Ontology Generation with Foundation Models](#).” In *International Conference on Very Large Data Bases (VLDB)* 2025.

3. **Sachin Konan**, Larry Rudolph, Scott Affens. "[Automating the Generation of Functional Semantic Types with Foundational Models](#)." In *The Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)* 2024.

2. **Sachin Konan**, Esmaeil Seraj, and Matthew Gombolay. "[Contrastive Decision Transformers](#)." In *International Conference on Robot Learning (CoRL)* 2023.

1. **Sachin Konan**, Esmaeil Seraj, and Matthew Gombolay. "[Iterated reasoning with mutual information in cooperative and byzantine decentralized teaming](#)." In *International Conference on Learning Representations (ICLR)* 2022.

PREPRINT/ARTICLES

2. Advait Gupta, **Sachin Konan**, Gabe Sarch, Zhuang Liu. "AgentVQA: A Unified Benchmark for Agentic Visual Understanding." In review at *International Conference on Learning Representations (ICLR)* 2026.

1. **Sachin Konan**, Kevin Liang, and Li Yin. "[Extending one-stage detection with open-world proposals](#)."

TEACHING

Georgia Institute of Technology - Teaching Assistant (with Constantinos Dovrolis)	Fall 2021 -
- Head Grader for CS 3510: Design and Analysis of Algorithms	Fall 2022

AWARDS

Princeton AI Lab Graduate Student Fellow	2025
Lambda AI Cloud Credits Grant Sponsorship	2025
NSF Graduate Research Fellowship (declined before deadline)	2022
PURA Undergraduate Research Fellowship	2021
NBA National Hackathon (Top 10 in the US)	2021
Davidson Fellow	2018
Finalist, Regeneron Science Talent Search	2018
2nd in Embedded Systems at Intel International Science and Engineering Fair	2017

COURSEWORK

Graduate-Level: COS 597D: Information theory in computer science, COS 597K: Frontiers in Deep Learning, CSCI-GA 3033: Deep Decision Making and Reinforcement Learning

Undergraduate Machine Learning: CS 4641: Machine Learning, CS 4803: Deep Learning, CS 4476: Computer Vision, CS 3630: Perception and Robotics, CS 3600: Intro To Artificial Intelligence

Undergraduate Computer Science: CS 4540: Advanced Algorithms, CS 3510: Design and Analysis of Algorithms, CS1302: Data Structures and Algorithms

Undergraduate Mathematics: MATH 3406: Advanced Linear Algebra, MATH 4032: Combinatorial Analysis, ISYE 3770: Statistics, Math 2550: Intro To Multivariable Calculus